

Advanced (Habanero)

- Q1.** Which lines are parallel given $y = 2x + 1$ and $y = 2x - 3$?
(A) The lines are parallel
(B) The lines are perpendicular
(C) The lines are skew
(D) The lines are not parallel
(E) Insufficient information
- Q2.** What is the slope of a line parallel to $y = 4x - 2$?
(A) 4
(B) -4
(C) 2
(D) -2
(E) 1
- Q3.** If $y = 3x + 1$ and $y = 3x - 2$, which statement is true?
(A) The lines intersect at $(0, 1)$
(B) The lines are parallel
(C) The lines are perpendicular
(D) The lines are skew
(E) The lines have the same y -intercept
- Q4.** What is the equation of a line parallel to $y = x - 3$ that passes through $(2, 5)$?
(A) $y = x + 3$
(B) $y = x - 1$
(C) $y = x + 1$
(D) $y = 2x - 3$
(E) $y = x + 5$
- Q5.** If two lines have the same slope, what can be concluded about the lines?
(A) The lines are parallel
(B) The lines are perpendicular
(C) The lines are skew
(D) The lines intersect at the origin
(E) The lines have the same y -intercept
- Q6.** Which of the following lines are parallel to $y = 2x + 1$ and $y = 2x - 3$?
(A) $y = x + 1$ and $y = x - 1$
(B) $y = 2x + 2$ and $y = 2x - 2$
(C) $y = 3x + 1$ and $y = 3x - 3$
(D) $y = x + 2$ and $y = x - 2$
(E) $y = 4x + 1$ and $y = 4x - 3$
- Q7.** If $y = mx + b$ is parallel to $y = 2x - 1$, what is the value of m ?
(A) 1
(B) 2
(C) -2
(D) -1
(E) 0
- Q8.** What is the relationship between the slopes of parallel lines?
(A) The slopes are equal
(B) The slopes are opposite reciprocals
(C) The slopes are negative reciprocals
(D) The slopes are perpendicular
(E) The slopes are skew
- Q9.** If two lines have the same slope, what can be concluded

Open Ended Questions (FRQ)

- Q9.** Find the equation of a line parallel to $y = 3x - 2$ that passes through $(1, 4)$
- Q10.** Prove that if two lines have the same slope, they are parallel

Chef's Tips

- Please ensure students have a graphing calculator
- Allow students to use a straightedge
- Remind students to show their work and explain their reasoning